Forging an Open Secure Control Plane for the AI Era

AI Open, secure, interoperable

Mamta Gupta
VP Compute & Communications Segments
LATTICE SEMICONDUCTOR

Kelly Bryant
VP Firmware and Solutions Strategy
ami™
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OPEN Compute Project®

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AI Redefines Infrastructure's Basic Unit

From servers to racks, clusters and AI factories managed as one system

100,000+	accelerators managed as a single training cluster
1 MW	per rack makes power and thermals a control problem
Heterogeneous	GPUs, CPUs, DPUs, NICs and custom silicon in every node

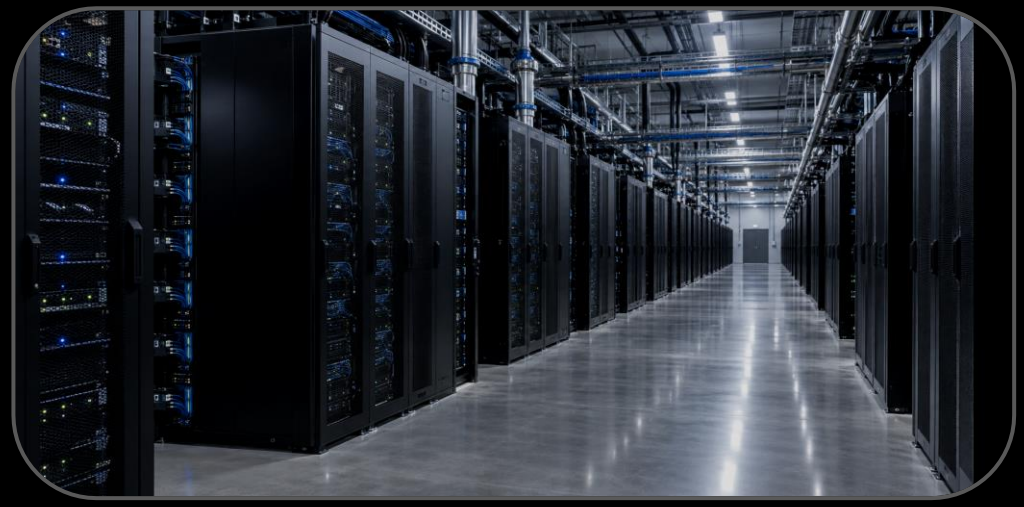
The bottleneck is shifting from compute to system operations

AI Rack Scale, Management Becomes System Control

SERVER-CENTRIC



SYSTEM SCALE



Manage the server

BMC • basic server lifecycle • device-level control

Control the system

Compute • power • cooling • network • lifecycle • security • telemetry

System control coordinates the functions that determine how the rack operates, responds and recovers

System Control Is a Cross-Layer Problem

OPERATORS	Run and deploy at scale
MANAGEMENT	Policy, lifecycle, and fleet state
FIRMWARE	Initialize, secure, and recover the platform
HARDWARE TRUST	Establish identity and integrity
SILICON	Compute, manage, and accelerate

IMPLICATION

If the layers operate independently, system-level control breaks down

The architecture must coordinate state, trust and action across the full stack.

Three Requirements Define the New Control Architecture



Secure

Trust rooted below the OS and attested continuously



Resilient

Detect, update, and recover without taking the cluster offline



Efficient

Use power, thermal, and health telemetry as a control loop

Secure, resilient, and efficient must be solved as one integrated system

An Open Control Plane Connects the Layers

Management & Orchestration	Fleet policy • telemetry • lifecycle
Platform Firmware & Control	Boot • update • recovery • power / thermal
Trust & Security	Identity • attestation • resilient recovery
Open Interfaces, Standards, & APIs	Interoperability across vendors and platforms

OPEN CONTROL PLANE

Connect these layers so heterogeneous AI infrastructure can operate as one secure, resilient system

Open Standards Create the Common Language

Management & Telemetry

Common models for monitoring, lifecycle and update

Identity & Attestation

Verify devices, measurements, and trust state

Root of Trust

Establish hardware-backed Identity and integrity

Validation & Assurance

Shared approaches for security and interoperability testing

These capabilities already exist across the ecosystem.
The opportunity is to connect them into one interoperable control model.

Open Integration Turns Standards Into Systems

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- Hardware root of trust
- Programmable platform control
- Telemetry & recovery

Ecosystem Partners

- Silicon & platform vendors
- ODMs / OEMs
- Operators & open communities

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- Platform firmware
- BMC / manageability
- Provisioning & lifecycle workflows

OPEN INTEGRATION + CO-VALIDATION

Across hardware, firmware, management, and platform control

What This Means for Customers

Adopt Faster

Build on common interfaces and reusable implementations

Deploy With Less Risk

Validate interoperability and security before production

Operate More Predictably

Consistent control behavior as system state changes

An open control plane helps AI infrastructure operate as one system without becoming another proprietary integration layer

The Pieces Exist. Now We Need to Connect Them.

An open control plane only works if the ecosystem aligns on the interfaces, trust model, and validation approach

CONVERGE	Common standards and profiles across platforms
ESTABLISH TRUST	Attestation measurement + open roots of trust
CONTRIBUTE	Upstream implementations and agreed upon standards
CO-VALIDATE	Interop testing across silicon, firmware and management